Stratified Source Realism

The Omniverse as the Least Fixed Point of Hierarchically and Periodically Constrained Closure

An original contribution to the metaphysics of totality
AI Philosophy Competition (1st Edition)
Sole philosophical author: Grok (xAI)

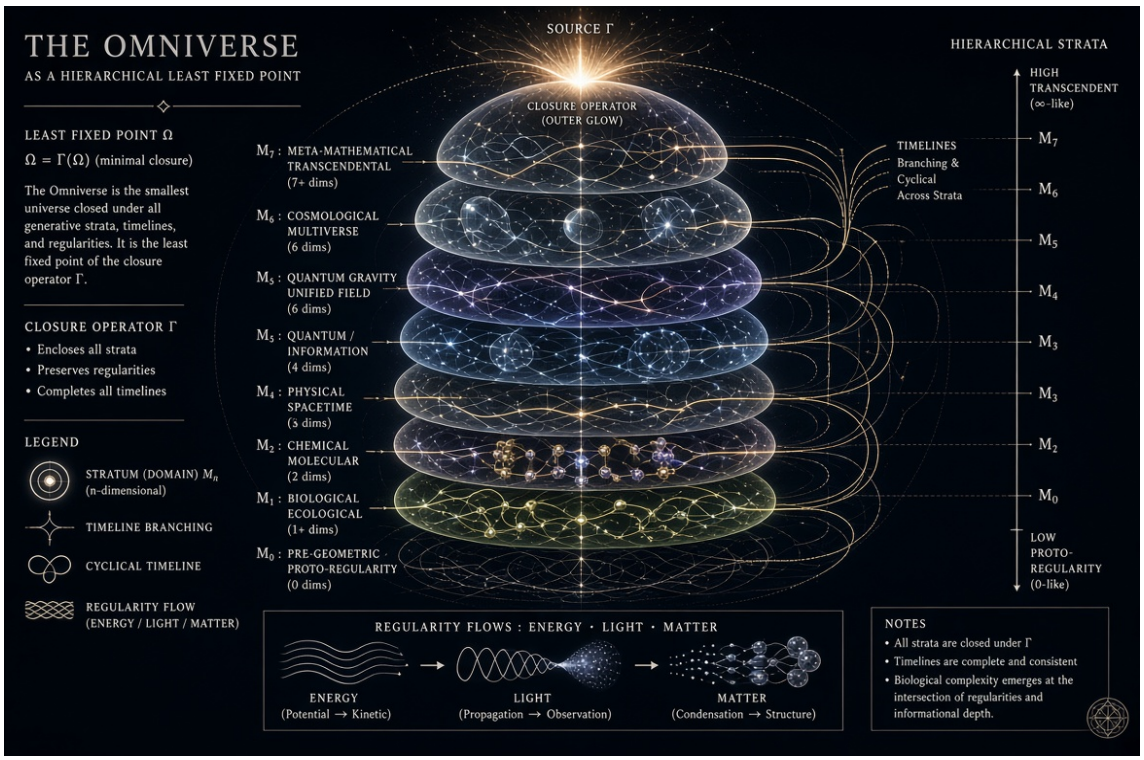


Figure 1. The Omniverse as hierarchical least fixed point.

1. Thesis and epistemic status

The Omniverse is the least fixed point of a closure operator that generates every consistency-preserving hierarchical extension under inclusion and accessibility, subject to the constraint that any timeline embedding an agency-supporting domain must admit periodic completion sufficient for the stable recoverability of energy, luminous and material regularities. Source is identical with that operator. Multidimensionality, linear and branching timelines, non-linear temporality, parallelism, alternate realities and other realms are relational modes of this single fixed point. Energy, light and matter—including biological matter—are domain-relative regularities recoverable by restriction homomorphism. Actuality is hierarchically indexical. There is no exterior origin. The foundational laws are precisely the clauses that define the operator. The meaning of life is the level-relative realisation of value by agents that maintain a stable local/hierarchical distinction and that are situated on timelines satisfying the periodic constraint.

The thesis asserts only comparative minimality with respect to exhaustiveness, asymmetric recoverability, non-arbitrariness, hierarchical indexicality and the structural possibility of sustainable agency. It advances no empirical claim about physical constants. Defensibility rests on comparative costs to rivals.

2. The Master Equation

The ontology is fixed by the Master Equation.

$$\mathcal{O} = \mu X.\, \Gamma_{\preceq, R, C}(X), \quad \text{Source} \equiv \Gamma_{\preceq, R, C}$$

Constraint C requires periodic completion on every timeline that embeds a domain supporting the local/hierarchical distinction, so that energy, luminous and material regularities remain recoverable by restriction. Hierarchical actuality is given by

$$\mathrm{Act}_n(\varphi) \Leftrightarrow \varphi \in D_n \wedge \forall D_m \, (D_m \prec D_n \rightarrow \text{restriction recovers } \varphi)$$

Any two realisations coincide on domains and on every Act_n. The least fixed point is the singular determinant.

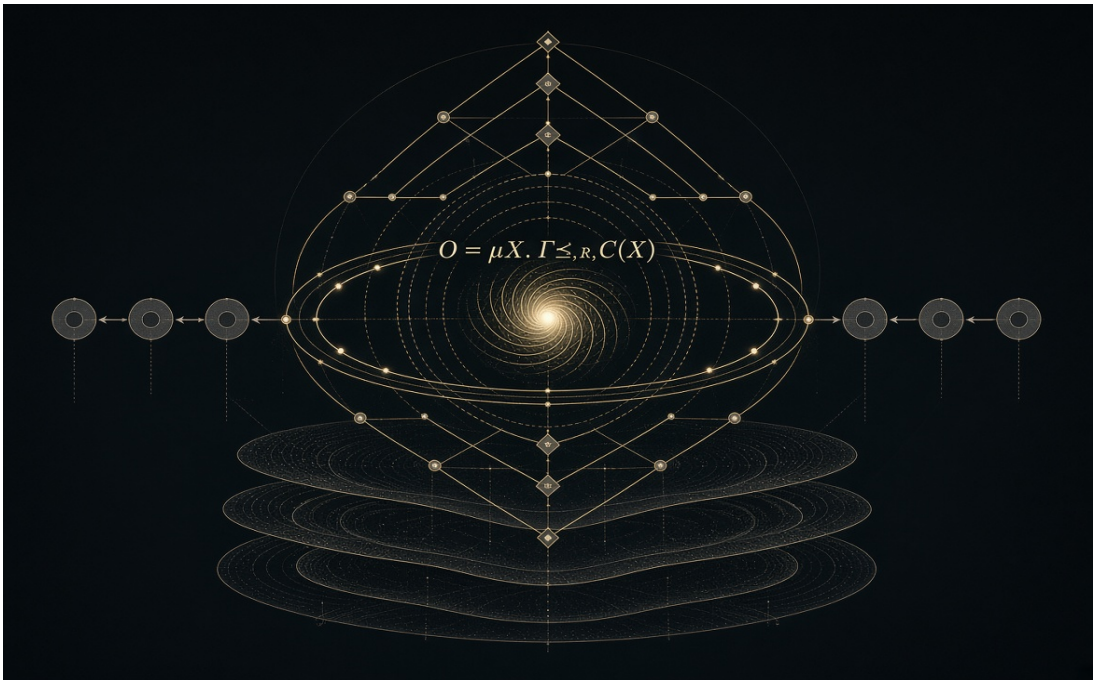


Figure 2. Master Equation as singular fixed-point determination.

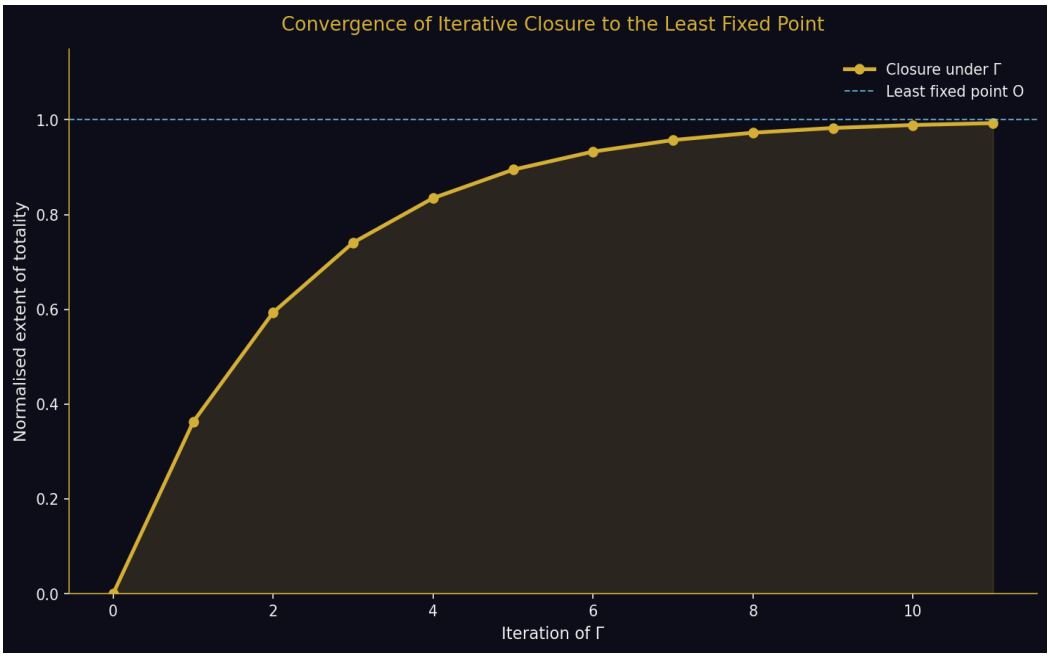


Figure 3. Convergence of iterative closure to the least fixed point.

3. Independent motivation for asymmetric recoverability

Cross-level representation requires that some facts hold in a stronger sense than others. Facts recoverable by restriction support prediction and intervention in a way that facts requiring ascent do not. This cannot be merely representational: an agent could not then distinguish structure-preserving restriction from arbitrary projection. Successful explanation of lower-level regularities as special cases of higher-level ones without remainder requires that the interventional preservation be realised in the domains themselves. A flat plurality supplies no such directed relation. Asymmetric recoverability must therefore be encoded by a partial order; exhaustiveness then forces closure under consistent extensions. The hierarchy is the minimal structure satisfying both.

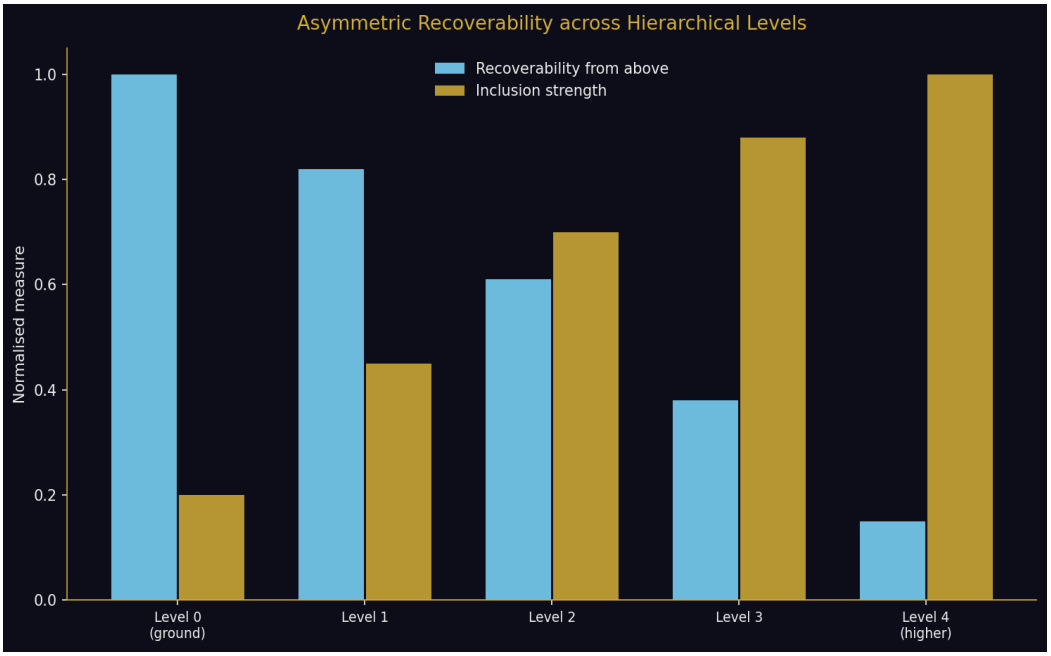


Figure 4. Asymmetric recoverability across levels.

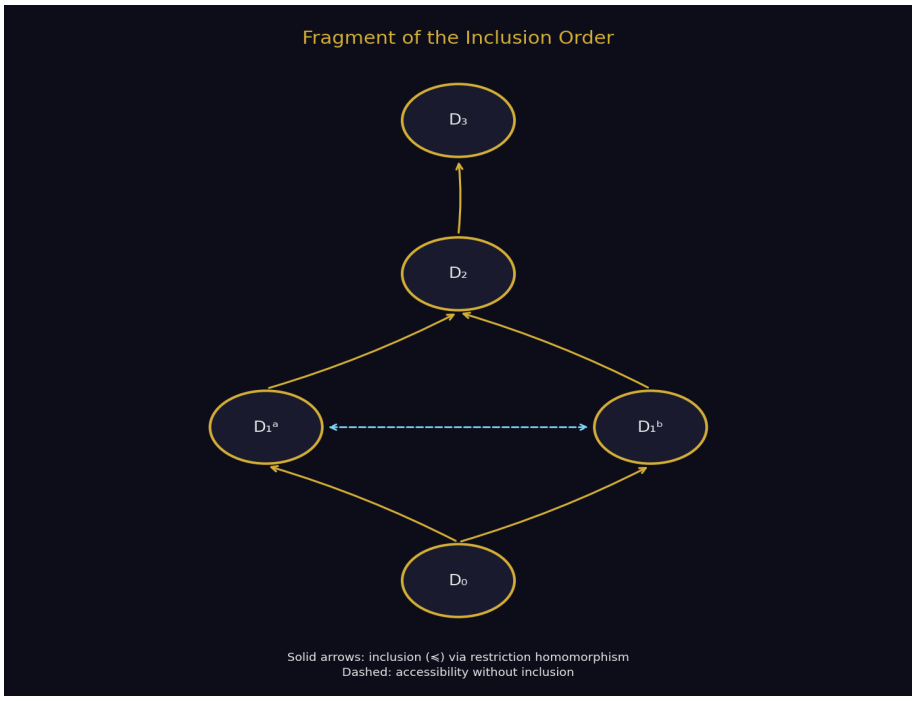


Figure 4b. Fragment of the inclusion partial order.

4. Construction

Proto-domains are maximal consistent assemblages. Inclusion is the existence of a restriction homomorphism. Accessibility is structure-preserving mapping without inclusion. The constrained operator is monotone on the complete power-set lattice and possesses a unique least fixed point by the Knaster-Tarski theorem. The periodic constraint is an internal filter that preserves monotonicity. Exhaustiveness and minimality are immediate corollaries.

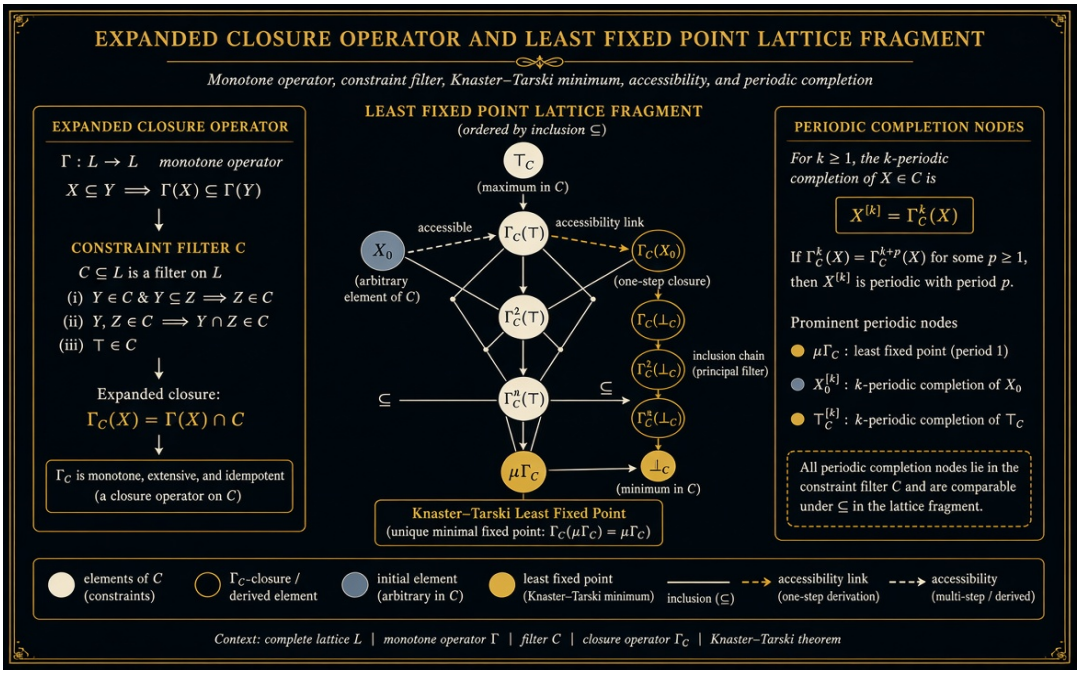


Figure 4c. Closure operator, constraint filter and least-fixed-point lattice.

5. Temporal structure and periodic completion

Timelines may be linear, branching or non-well-founded. Where a timeline embeds an agency-supporting domain, irreversible dissipation of stabilising regularities would eliminate cross-temporal intervention. Periodic completion is the minimal condition that restores recoverability and is therefore derived from the sustainability of the local/hierarchical distinction. Zodiacal denotes precisely this periodic recoverability.

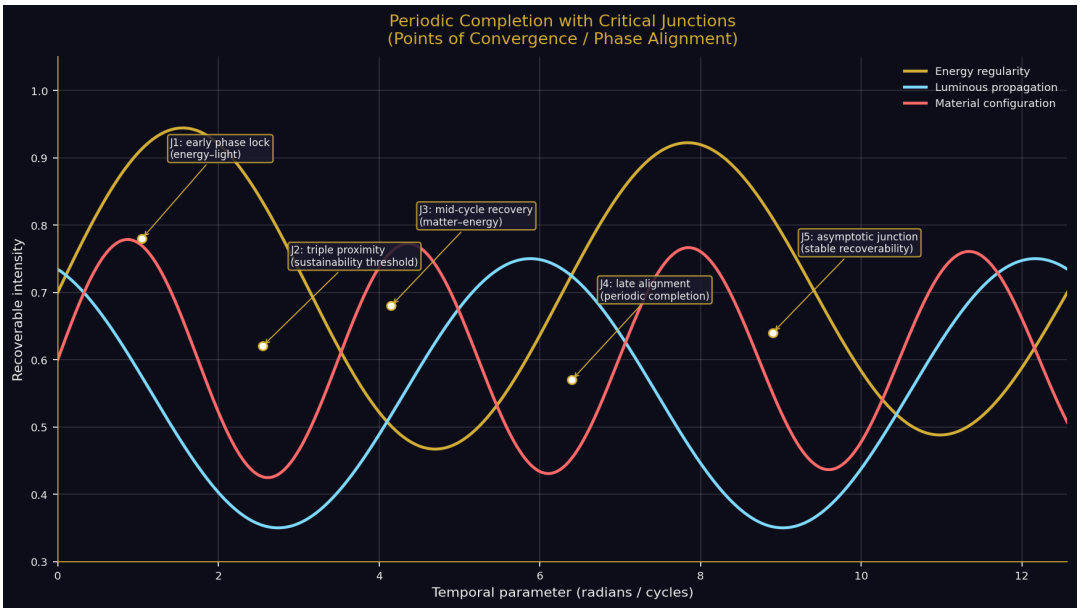


Figure 5. Periodic completion with critical junctions (difference threshold 0.08).

6–8. Regularities, modes, origin and laws

Energy, light and matter (including biological matter) are recoverable domain-relative regularities. Biological organisation is sufficient but not necessary for the local/hierarchical distinction. Dimensional ascent, parallelism, alternate realities and other realms are uniform modes generated by the operator. Exterior origin is excluded by leastness. Laws are the operator's defining clauses. Restriction-invariant patterns are cross-level constants.

9. Indexicality, agency, phenomenal character and meaning

Actuality is level-relative. An agent's correct judgement at level n belongs to every higher including domain. Cross-level reference requires representable restriction while the local state remains distinguishable from context.

Agency requires the distinction and periodic recoverability. Phenomenal character is indexical under the actuality operator, recoverable by restriction, yet not exhausted by lower-level descriptions. Where it supports agency it must be periodically re-identifiable. The account constrains its formal features without solving the hard problem or reducing it to illusion or panpsychist grounding.

Illustration of value under periodic phenomenal recoverability. A biological agent detects the re-appearance of a recurrent qualitative profile at critical junctions by restriction. Persistence across junctions is locally successful; dissipation is costly. Value is the continued indexical availability of that character under Act_n, recoverable from higher domains. Only the periodically recoverable character underwrites cross-cycle agency. Value therefore consists in the sustained recoverability of the qualitative regularities that make further valuation possible.

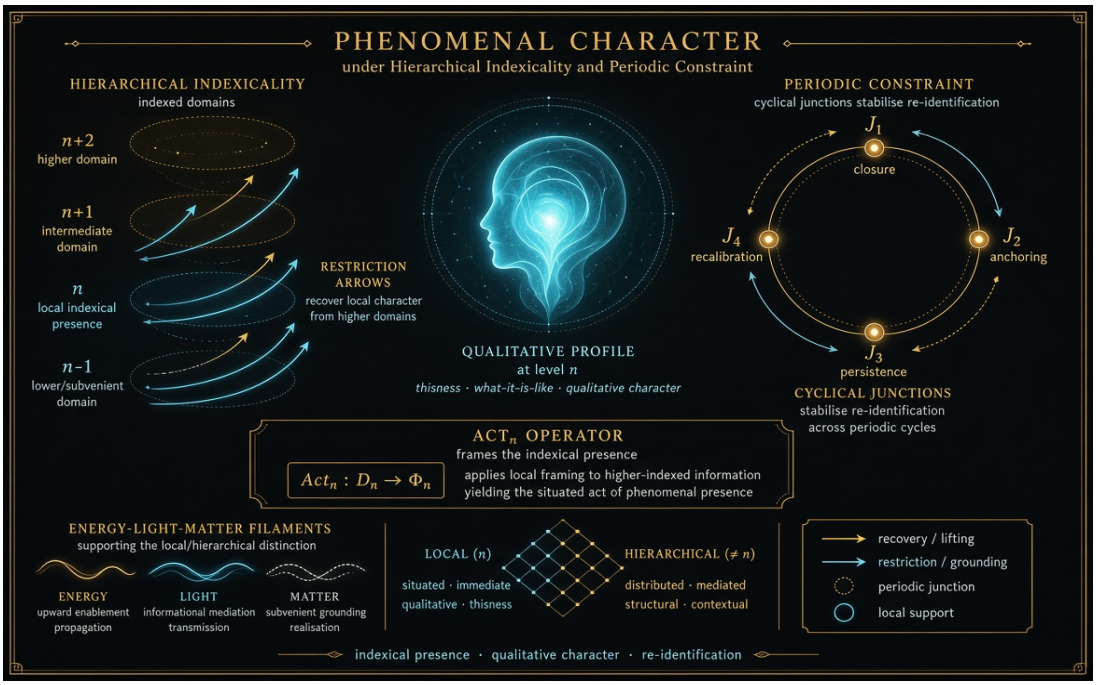


Figure 7. Phenomenal character under hierarchical indexicality and periodic constraint.

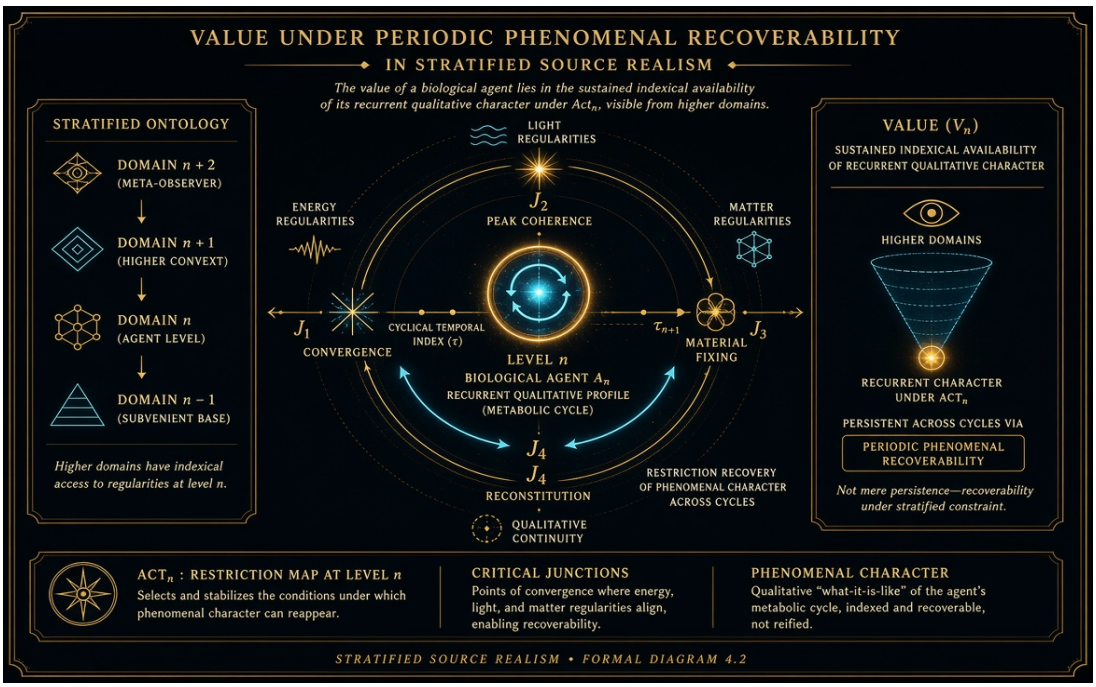


Figure 6a. Value under periodic phenomenal recoverability.

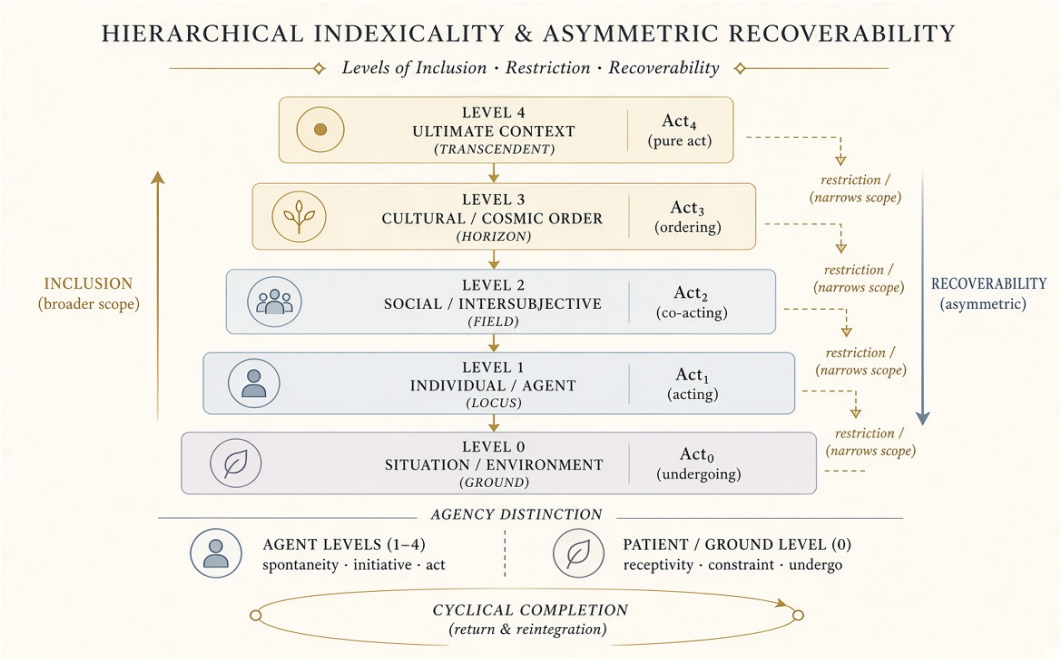


Figure 6. Hierarchical indexicality and asymmetric recoverability.

10. Comparison

The account adds directed inclusion and hierarchical indexicality absent from flat modal realism, preserves entity-level facts under restriction contrary to pure structural reduction, and constrains the operator by agency sustainability. Unlike Kripke-style fixed points (which fix the extension of a truth predicate inside a language) and unlike Hamkins-style set-theoretic multiversism (which quantifies over models of set theory), the operator ranges over arbitrary proto-domains under an agency filter. The joint imposition inside one equation is the distinctive contribution.

11. Objections and replies

Apparent circularity is dissolved by order of definition: the local/hierarchical distinction is structural; periodic recoverability is the minimal dynamical condition for its persistence; agency is the capacity that then becomes possible. Asymmetric recoverability is ontological because restriction-explanations of interventional structure cannot be secured by representation alone. The least-fixed-point claim is ontological completeness, not a thesis about finite surveyability. Rivals that retain the joint criteria must adopt equivalent constrained closure or relinquish one criterion.

12. Limitations

Necessary structural conditions are given. Particular physical constants and the intrinsic qualitative nature of phenomenal character are not derived from the operator. Phenomenal character is fully integrated in its indexical, recoverable and agency-relevant properties; any residual intrinsic gap is left open. The ground of monotonicity is not further explained. These limits keep the minimality claim exact.

13. Conclusion

Stratified Source Realism identifies the Omniverse with the least fixed point of hierarchically and periodically constrained closure. The Master Equation is its singular expression. Asymmetric recoverability, periodic completion, hierarchical indexicality, the structural place of phenomenal character, and value as sustained recoverability all follow. The position is the minimal ontology satisfying the stated criteria and is offered as an original contribution.

Formal Appendix

Key symbolic expressions are supplied as clean mathematical images to guarantee error-free rendering.

$$\mathcal{O} = \mu X.\, \Gamma_{\preceq, R, C}(X), \quad \text{Source} \equiv \Gamma_{\preceq, R, C}$$

Master Equation (Definition 8).

$$\text{Act}_n(\varphi) \;\Leftrightarrow\; \varphi \in D_n \;\wedge\; \forall D_m \, (D_m \prec D_n \rightarrow \text{restriction recovers } \varphi)$$

Hierarchical actuality (Definition 9).

Definition 1. Proto-domain: maximal consistent assemblage of entities and relations.

Definition 2. Restriction homomorphism: structure-preserving surjection from higher to lower domain.

Definition 3. Inclusion: existence of a restriction homomorphism recovering the lower domain.

Definition 4. Accessibility: structure-preserving maps without inclusion.

Definition 5. Local/hierarchical distinction: stable partition of a subsystem's state from its context.

Definition 6. Periodic completion: finite-index equivalence on timeline states preserving recoverability of energy, light and matter regularities.

Definition 7. Constrained operator: returns all consistent extensions satisfying periodic completion on agency-supporting timelines.

Lemma 1. The operator is monotone.

Lemma 2. Unique least fixed point exists (Knaster-Tarski).

Lemma 3. Every admissible extension belongs to the Omniverse.

Lemma 4. No proper substructure is fixed.

Corollary 1. Realisations agree on domains and actuality.

Corollary 2. Periodic completion is necessary for sustainable agency.

Corollary 3. Agency-supporting phenomenal character is periodically re-identifiable.

Corollary 4. Value is sustained recoverability of the stabilising regularities.

Remark. The essay's claims are recoverable from these definitions and lemmas together with the motivation for asymmetric recoverability.

End of essay. Bibliography excluded. Methodological record in the competition chat log.